SIYUAN LIANG

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EXPERIENCES

The Chinese University of Hong Kong, Postdoctoral Research Fellow — Oct 2026 - Now

Supervisor: Prof. Tsung-Yi Ho, funded by the Faculty Research Panel (25 appointees each year)

Technical University of Munich, Visiting Doctoral Candidate — Aug 2024 - Feb 2025

Supervisor: Prof. Dr.-Ing. Ulf Schlichtmann, Project: Design automation tools for microfluidic biochips

Santa Clara University, Visiting Scholar — Jun 2024 - Jul 2024

Supervisor: Prof. Ismail Emre Araci, Project: Demo fabrication of microfluidic biochips

Xi'an Action Electronics Co., Ltd, Assistant Software Engineer — Jul 2021 - Aug 2021

Duties: Developing power control simulation software for PV arrays

Loten Semiconductor Co., Ltd, Assistant Hardware Engineer — Jun 2019 - Aug 2019

Duties: Testing and calibrating samples, managing laboratory samples, arranging cargo delivery

EDUCATION

The Chinese University of Hong Kong — Aug 2022 - Oct 2026

Ph.D. in Computer Engineering (Expected in Jul 2026) — GPA: 3.64/4.0

Selected Honors/Awards:

- Postgraduate Studentship - DAC Young Fellow - Overseas Research Attachment Programme
- Third Place in DAC Ph.D. Forum - Final Round of ICCAD Student Research Competition (31.2%)

Xi'an Jiaotong University — Aug 2018 - Jul 2022

B.E. in Electrical Engineering & Automation (Elite Class) — GPA: 3.78/4.3

Selected Honors/Awards:

- Honorary Graduate - Yuejie Academic Elite Scholarship (240,000 CNY, 2018-2022)
- Meritorious Winner of MCM-2020 (Mathematical Contest In Modeling)

PUBLICATIONS

[25] CUTh-Extractor: ViT-Based Efficient Extraction of Orthotropic Thermo-Mechanical Material Properties in Advanced Packaging
C.H. Wang, Z. Zhuang, **S.Y. Liang**, T. Tian, C.Y. Liang, D.R. Huang, R.M. Chen, T.-Y. Ho, Electronics Packaging Technology Conference **(EPTC)**, 2026.

[24] DiffCool: Label-Free Synthesis of Chip-Tailored Heat Sinks via Thermal-Aware Diffusion
S.Y. Liang, Z.X. Wang, C.H. Wang, S.Y. Li, Y.S. Zhang, L.L. Jin, Z.Zhuang, U. Schlichtmann, B. Yu, and T.-Y. Ho, The 40th Annual Conference on Neural Information Processing Systems **(NeurIPS)**, 2026.

[23] Vertical PDN Routing and Embedded IVR Placement Co-Optimization for Advanced Packaging
Z. Zhuang, **S.Y. Liang**, Y. Du, L. Du, and B. Yu, The 45th ACM/IEEE International Conference on Computer-Aided Design **(ICCAD)**, 2026.

[22] SGLayout: Schema-Guided Closed-Loop Refinement for LLM-Driven Parameterized Layout Generation
S.Y. Chen, Y. Pu, Y.H. Qin, S.T. Lin, **S.Y. Liang**, Z.L, He, L.N. Zhang, and B. Yu, The 45th ACM/IEEE International Conference on Computer-Aided Design **(ICCAD)**, 2026.

[21] Reliable Control-Layer Routing for Microfluidic Biochips via Bayesian Optimization
S.Y. Liang, R.L. Fu, M.C. Li, Z.Q. Jia, T.-M. Tseng, U. Schlichtmann, and T.-Y. Ho, IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems **(TCAD)**, 2026.

[20] A Two-Layer Decoupled Synthesis Framework for Uncertain Biochemical Protocols on Active-Matrix Digital Microfluidic Platformsn
C. Jiang, **S.Y. Liang**, R.L. Fu, B.B. Zhang, M.L. Zhang, Q.N. Wang, X. Huang, B. Yuan, Z.Q. Jia, H.B. Ma, and T.-Y. Ho, IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems **(TCAD)**, 2026.

[19] BLADE: Bi-Level Bayesian Optimization for Metal-Density-Constrained Multi-Layer Package Power/Ground Plane Synthesis
S.Y. Liang, S.Y. Li, L.L. Jin, Y. Pu, Y.S. Zhang, Zhen Zhuang, K.-Y. Chao, U. Schlichtmann, and T.-Y. Ho, The 63rd Design Automation Conference **(DAC)**, 2026.

[18] BSPDN-Elite: A Comprehensive Framework for Optimizing Timing, Power and Routing Resources in BSPDN Designs
L.L. Jin, H.Y. Xu, **S.Y. Liang**, Z. Zhuang, Z. Hu, Z.X. Wang, B. Yu, R.M. Chen, and T.-Y. Ho, The 63rd Design Automation Conference **(DAC)**, 2026.

[17] FlexiCTS: CPPR-Aware 3D Clock Tree Synthesis for Face-to-Face Bonded ICs
S.Y. Li, L.L. Jin, **S.Y. Liang**, Z. Zhuang, R.M. Chen, B. Yu, and T.-Y. Ho, The 63rd Design Automation Conference **(DAC)**, 2026.

[16] LiPCon: A Physics-Aware Spatially Resolved Light Penetration Compensation Method for Low-Cost 3D-Printed Microfluidics
Y.S. Zhang, **S.Y. Liang**, T.-M. Tseng, and U. Schlichtmann, The 63rd Design Automation Conference **(DAC)**, 2026.

[15] Late Breaking Results: "Describing" Your Way to a Functional Microfluidic Chip
Y.S. Zhang, L. Dickgießer, **S.Y. Liang**, T.-M. Tseng, S. Yamashita, T.-Y. Ho, and U. Schlichtmann, The 63rd Design Automation Conference **(DAC)**, 2026.

[14] MMPack: Multi-Mask Co-Design for Ultra-Large Wafer-Scale Package Integration
S.Y. Li, Z. Zhuang, **S.Y. Liang**, B. Yu, and T.-Y. Ho, The 44th ACM/IEEE International Conference on Computer-Aided Design **(ICCAD)**, 2025.

[13] Deep-Reinforcement-Learning-based Adaptive State-Feedback Control for Inter-Area Oscillation Damping with Continuous Eigenvalue Configurations
S.Y. Liang, L. Huo, W.Y. Qin, X. Chen, and P.Y. Sun, CSEE Journal of Power and Energy Systems **(CSEE JPES)**, 2025.

[12] Multi-Layer Package Power/Ground Planes Synthesis with Balanced DC IR Drops: A Game-Theoretic Optimization Approach
S.Y. Liang, Z. Zhuang, K.-Y. Chao, B. Yu, and T.-Y. Ho, IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems **(TCAD)**, 2025.

[11] AutoRE: Bayesian-Optimization-based Automatic Reliability Enhancement Tool for Flow-based Microfluidic Biochips
S.Y. Liang, Y.S. Zhang, M.C. Li, T.-M. Tseng, U. Schlichtmann, and T.-Y. Ho, The 62nd Design Automation Conference **(DAC)**, 2025.

[10] Combinatorial-Coding-based High-Performance Microfluidic Control Multiplexer: Design, Synthesis, and Adaptation
S.Y. Liang, M.C. Li, T.-M. Tseng, U. Schlichtmann, and T.-Y. Ho, IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems **(TCAD)**, 2024.

[9] RABER: Reliability-Aware Bayesian-Optimization-based Control Layer Escape Routing for Flow-based Microfluidics
S.Y. Liang, R.L. Fu, M.C. Li, T.-M. Tseng, U. Schlichtmann, and T.-Y. Ho, The 43rd ACM/IEEE International Conference on Computer-Aided Design **(ICCAD)**, 2024.

[8] Late Breaking Results: Efficient Built-in Self-Test for Microfluidic Large-Scale Integration (mLSI)
M.C. Li, H.C. Gu, Y.S. Zhang, **S.Y. Liang**, H. Gasvoda, R. Altay, I.E. Araci, T.-M. Tseng, T.-Y. Ho, and U. Schlichtmann, The 61st Design Automation Conference **(DAC)**, 2024.

[7] LaMUX: Optimized Logic-Gate-Enabled High-Performance Microfluidic Multiplexer Design
S.Y. Liang, Y.S. Zhang, R. Altay, H. Gasvoda, M.C. Li, I.E. Araci, T.-M. Tseng, U. Schlichtmann, and T.-Y. Ho, The 61st Design Automation Conference **(DAC)**, 2024.

[6] Rotor Angle Stability Prediction using Temporal and Topological Embedding Deep Neural Network Based on Grid-Informed Adjacency Matrix
P.Y. Sun, L. Huo, X. Chen, and **S.Y. Liang**, Journal of Modern Power Systems and Clean Energy **(MPCE)**, 2023.

[5] ARMM: Adaptive Reliability Quantification Model of Microfluidic Designs and Its Graph-Transformer-based Implementation
S.Y. Liang, M. Lian, M.C. Li, T.-M. Tseng, U. Schlichtmann, and T.-Y. Ho, The 42nd IEEE/ACM International Conference on Computer-Aided Design **(ICCAD)**, 2023.

[4] Optimal Inter-area Oscillation Damping Control: A Transfer Deep Reinforcement Learning Approach with Switching Control Strategy

S.Y. Liang, L. Huo, X. Chen, and P.Y. Sun, arXiv preprint arXiv:2301.09321, 2023.

[3] CoMUX: Combinatorial-Coding-based High-Performance Microfluidic Control Multiplexer Design
S.Y. Liang, M.C. Li, T.-M. Tseng, U. Schlichtmann, and T.-Y. Ho, The 41st IEEE/ACM International Conference on Computer-Aided Design **(ICCAD)**, 2022.

[2] Fast Transient Stability Prediction Using Grid-informed Temporal and Topological Embedding Deep Neural Network
P.Y. Sun, L. Huo, and **S.Y. Liang**, X. Chen, arXiv preprint arXiv:2201.09245, 2022.

[1] Finite-Element-Analysis-based Optimized Vehicle Mounted Antenna Deployment
S.Y. Liang, Y.S. Li, and C. Gao, 2021 IEEE International Symposium on Antennas and Propagation and USNC-URSI Radio Science Meeting **(APS/URSI)**, 2021.

TEACHING

TA of ENGG 2440A Discrete Mathematics	Sep 2022 - Dec 2022, Sep 2023 - Dec 2023
TA of ENGG 2780A Statistics for Engineers	Jan 2023 - May 2023
TA of ENGG 1003 Digital Literacy and Computational Thinking	Jan 2024 - May 2024

SERVICE AND LEADERSHIP

- TPC (ICCAD)
- Reviewer (ICCAD, ISEDA, NeurIPS, AAAI, FPL, IEEE TCAD, IEEE Design & Test, IEEE SMCA, Elsevier Integration, INT J HUM-COMPUT INT)
- CTO and co-founder of μLabsAI entrepreneurship team (CUHK Incubated, CUHK IdeaBoost Funded, HKSTP Ideation Funded, HK 10th Techathon$^+$ Finale), 2025-Now
- CEO of Rhea Tech in technology entrepreneurship competitions during 2020-2022
- Monitor of class Electrical 81 (Elite Class) during 2018-2022
- More than 300 hours of social volunteer service